



# Legislative Consensus Engine



## Admin Portal

*A dual-sided legislative mediation platform*

and publish interactive polls.



## Voter Portal

Review simplified clauses, cast your vote, and instantly  
view the friction map.

Built by Sarthak Kastiya

# Democracy has a precision problem

Bills are written in legal language the public cannot read

Existing tools only capture Yes or No on a whole bill

Legislators cannot identify which specific clause is causing friction

Citizens have no structured way to express nuanced opinion on legislation

WHO IS THIS FOR?

# Who are you building this for and why do they need it?



## LEGISLATORS

Need clause-level public sentiment data before a vote — not after.

Currently make decisions without knowing where exactly the public is pushing back.

The tool gives them a Friction Map: a ranked list of every clause by opposition score.



## CITIZENS

Need to understand what they are actually voting on.

Both sides of every argument are presented fairly — informed choice, not gut reaction.

One platform that translates legal text into plain language they can act on.

# How it works

1

## UPLOAD

Admin uploads a PDF or pastes raw bill text

2

## EXTRACT

AI extracts every clause and writes a plain-language summary

3

## ARGUE

One Argument For and one Argument Against is generated per clause

4

## VOTE

Citizens vote Yes, Neutral, or No — clause by clause

[← Back to Dashboard](#)

## Create New Poll

Upload raw legislative text or a PDF to generate a consensus poll using AI.



Upload PDF Bill

Drag and drop or click to browse

OR PASTE TEXT

Paste raw legislative text or demo triggers here...

▷ GENERATE POLL QUESTIONS

# Pure Democratic Math

## Formula:

$$S_{\text{final}} = \Sigma(\text{votes}) / N$$

## Scale:

+10 Approve | 0 Neutral | -10 Oppose

## Bias:

Zero algorithmic intervention

## Equality:

Every human voice — equal weight

## The Consensus Algorithm



### The Consensus Algorithm

We utilize a strict, unweighted mathematical average to quantify democratic agreement. The system guarantees that 100% of the final consensus is driven by actual human voters, ensuring complete algorithmic neutrality and absolute fairness.

$$S_{\text{final}} = (\text{Sum of all human votes}) / (\text{Total number of voters})$$

#### 1. THE VOTING SCALE

Every participant casts a numerical value for each legislative clause: **+10** (Strongly Approve), **0** (Neutral), down to **-10** (Strongly Oppose).

#### 2. ABSOLUTE EQUALITY

The final consensus score is the absolute, unweighted average of all human votes cast. There is zero algorithmic intervention, ensuring every human voice holds the exact same mathematical power.

# How it is built

## Frontend

React 19 + Vite — component-based, optimized production builds

## Styling

Tailwind CSS — custom dark-mode design system

## PDF Parsing

PDF.js — full in-browser extraction, file never leaves device

## Prompt Design

Both arguments in a single AI call — prevents directional bias

## Deployment

Vercel — environment-variable-protected API keys

## AI FAILOVER CHAIN

### PRIMARY

**Gemini 1.5 Flash**

Speed-optimized

### SECONDARY

**GPT-4o-mini**

Precision fallback

### TERTIARY

**Llama 3.3-70B**

via Groq — low-latency

*Scoring Engine: Pure arithmetic mean — fully decoupled from AI*

# How does this help people rather than decide for them?



## AI Translates, Never Interprets

The AI converts legal text to plain language — it does not recommend, judge, or express an opinion.



## Both Sides, Always

One Argument For and one Argument Against are shown before the citizen votes. Informed choice, not gut reaction.



## 100% Human-Driven Score

The final score is pure arithmetic mean of human votes. AI cannot influence the result.



## The Tool Surfaces the Map

It shows where friction is. Humans navigate. Legislators decide.

***"The AI translates. The humans decide."***

# Real votes. Real friction. Real clarity.

 TOTAL VOTES CAST

1001

 FINAL BILL SCORE

-0.46

Public broadly against this bill

 MOST CONTENTIOUS

Nullification of Existing State Quotas

Rating: -6.4

## Clause Breakdown

| CLAUSE                          | VOTES (Y/N/O)   | FINAL SCORE | ZONE                       |
|---------------------------------|-----------------|-------------|----------------------------|
| The Trade and Commerce Act      | 49 / 178 / 773  | -7.2        | RED                        |
| Essential Commodities Amendment | 28 / 220 / 752  | -7.2        | RED                        |
| Contract Farming Framework      | 133 / 196 / 671 | -5.4        | 6 of 7 clauses in friction |
| Dispute Resolution Mechanism    | 130 / 194 / 676 | -5.5        |                            |
| Electronic Trading Platforms    | 482 / 204 / 314 | 1.7         |                            |
| Abolition of APMC Mandi Tax     | 200 / 194 / 606 | -4.1        | RED                        |
| Exemption from State Laws       | 75 / 204 / 721  |             |                            |

Only clause with consensus — legislator knows exactly where to focus



# Scale and real-world applications



## Protests & Negotiations

Identify which specific clauses drive opposition. Gives negotiators a precise starting point instead of a general grievance.



## Legislative Assembly Debates

Use the Friction Map to guide debate order. Start with Red Zone clauses — make parliamentary time evidence-based.



## Local Governance

Ward-level, constituency-level, or departmental consultations before bills reach the national level.



## Any Governing Body, Any Scale

Same platform for 50 people in a local council or 50,000 in a national consultation.

# What stronger AI guardrails look like



## Human Moderation Layer

Admins review and approve every AI-generated argument before voters see it



## Adversarial Red-Teaming

A second AI model reviews the first model's arguments for imbalance before publishing



## Rate-Limited Voting

Session validation and rate limits to prevent bot-driven vote manipulation



## Confidence Scoring

AI flags low-confidence clauses for expert human review before publishing



## Source Citation Requirement

AI must cite the exact original clause text it summarised — every output is auditable



## Immutable Audit Trail

All AI outputs stored alongside original bill text for independent post-consultation review

# Where this goes next



## Multi-Language Support

AI translates legal text and generates arguments in the local language — any country, any legislature



## Digital Identity Integration

Tie votes to verified national IDs — data credible enough to formally inform a parliamentary vote



## Live Real-Time Sessions

Friction Map updates live on a projector as thousands vote from their phones in a public town hall



## International Treaties

Cross-border documents consulted across multiple populations simultaneously



## Corporate Governance

Shareholder resolutions, board-level policies, internal company bylaws



## Government API

Any national digital platform integrates clause-level voting natively without building from scratch



# Consensus Engine

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A dual-sided legislative mediation platform.

# Legislative Consensus Engine

*"The AI translates. The humans decide."*

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Upload raw legislation  
and public comments

Live at: [legislative-consensus-engine.vercel.app](https://legislative-consensus-engine.vercel.app)

Share your vote, and instantly  
see the consensus map

[github.com/Sarthak9Kastiya/legislative-consensus-engine](https://github.com/Sarthak9Kastiya/legislative-consensus-engine)

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